

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Coding Challenge Task

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO4: Articulation (BL4, BL5)	Assessment:	Assessment Tool 1 – Coding Challenge Task
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO4 Statement:	Solve optimization problems using dynamic programming and greedy strategies.		
Student Name:	Shaik Nuha	Reg. No.:	192472242
Section / Batch:	CSE-AI	Date:	

Instructions to Students

- Answer both questions. Each question carries 50 Marks. Total: 100 Marks.
- Carefully analyze the given questions.
- Write clear code for the given problem.
- Analyze the Time complexity and Space complexity of your code using dynamic programming and greedy strategies

Question Type	Description	Questions	Total Marks
Coding Challenge Task	Focus on Code and analyze optimization problems using dynamic programming and greedy strategies	Q1 – Q2	100

SECTION A – Coding Challenge Task

Q70. Minimum Insertions to Balance Parentheses

Given a string of parentheses, determine the minimum number of insertions required to make it valid. The objective is to balance the parentheses. Efficient counting is required. Use Greedy or Dynamic Programming.

Input Format:

s = parentheses string

Sample Input:

s = (()

Expected Output:

Min Insertions = 1

Q71. Maximum Profit with Cooldown

Given stock prices, determine the maximum profit where after selling a stock, a cooldown of one day is required before buying again. The objective is to maximize profit under constraints. Use Dynamic Programming.

Input Format:

prices[]

Sample Input:

prices = 1 2 3 0 2

Expected Output:

Max Profit = 3

Code of Conduct

I Shaik Nuha(192472242) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Coding Challenge Task

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly understands problem constraints, edge cases, and competitive requirements	Understands problem with minor gaps in constraints	Partial understanding; misses key constraints	Misinterprets problem or constraints
Algorithm	25%	Selects optimal algorithm with best time and space complexity for competitive constraints	Uses efficient algorithm with minor scope for improvement	Uses basic/inefficient approach	No clear algorithmic approach
Code Implementation	20%	Writes correct, efficient, and clean code that passes all constraints	Code works with minor errors or inefficiencies	Partially correct code; fails for some cases	Code does not execute or gives incorrect results
Competitive Performance	20%	Solves problem within time limits and handles large inputs effectively	Solves within acceptable time with minor delays	Struggles with time limits or larger inputs	Unable to solve within time constraints
Test Case Handling	15%	Handles all test cases including edge and hidden cases	Handles most test cases	Misses important edge cases	Fails on multiple test cases

Mark Evaluation:

Questions	Problem Understanding (20)	Algorithm (25)	Code Implementation (20)	Competitive Performance (20)	Test Case Handling (15)	Total Marks (Each - 50)
Q1						
Q2						
Overall Total						

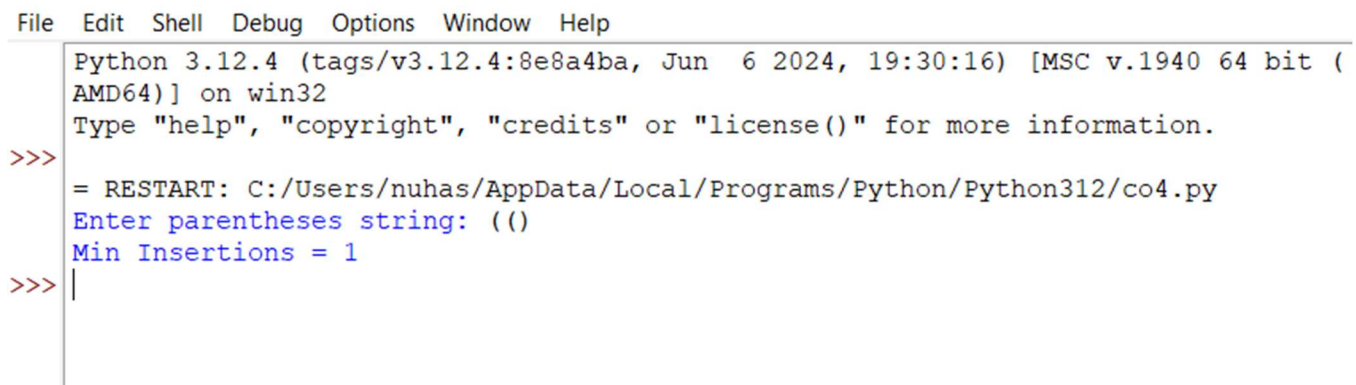
Faculty In-charge	Course Coordinator	Head of Department
Name: _____ Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

```

Q1. def min_insertions(s):
    balance = 0
    insertions = 0
    for ch in s:
        if ch == '(':
            balance += 1
        else:
            if balance > 0:
                balance -= 1
            else:
                insertions += 1
    return insertions + balance

s = input("Enter parentheses string: ")
print("Min Insertions =", min_insertions(s))

```



The screenshot shows a Python IDE window with the following content:

```

File Edit Shell Debug Options Window Help
Python 3.12.4 (tags/v3.12.4:8e8a4ba, Jun 6 2024, 19:30:16) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/nuhas/AppData/Local/Programs/Python/Python312/co4.py
Enter parentheses string: (()
Min Insertions = 1
>>>

```

```

Q2. def max_profit(prices):
    if not prices:
        return 0
    hold = -prices[0]
    sold = 0
    rest = 0
    for i in range(1, len(prices)):
        prev_sold = sold
        sold = hold + prices[i]
        hold = max(hold, rest - prices[i])
        rest = max(rest, prev_sold)

```

```
    return max(sold, rest)
prices = list(map(int, input("Enter prices: ").split()))
print("Max Profit =", max_profit(prices))
```

```
==== RESTART: C:/Users/nuhas/AppData/Local/Programs/Python/Python312/co4.py ====
Enter prices: 1 2 3 0 2
Max Profit = 3
>>> |
```